Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	Individual
Project Name	EV Market Performance & Sustainability Analytics
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Connect the supplied EV CSV/Excel dataset to Tableau and access fields for sales, category, model year, country, manufacturer, battery, price, range, charging, efficiency, ratings, and CO2 savings.	High
2	Data Processing	Verify field data types, handle missing values where appropriate, check duplicate records, standardize category and battery labels, and ensure units are consistent.	High
3	Business Analysis	Calculate total sales, average vehicle price, average driving range, annual CO2 saved, category sales share, battery-type sales share, and year-wise sales trends.	High
4	Business Insights & Analytical Outputs	Provide KPI cards, a sales trend line chart, country sales bar chart, category donut chart, battery technology donut chart, heatmap, manufacturer comparison, range distribution, and charging-efficiency scatter plot.	High
5	User Interaction & Analysis	Provide filters for model year, category, and battery type, plus tooltips and cross-chart exploration where configured.	Medium
6	Data Update & Reporting	Allow the workbook to refresh when the source dataset is updated and support export or publishing through available Tableau features.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Use clear titles, readable labels, consistent colours, and a simple layout so users can understand the dashboard with minimal training.
NFR-2	**Security**	Use approved access permissions for the workbook and source data; do not publish confidential or personally identifiable information publicly.
NFR-3	**Reliability**	Validate data types, calculations, and aggregation levels so KPIs and charts consistently represent the intended measures.
NFR-4	**Performance**	Use efficient calculations, avoid unnecessary marks and filters, and keep the dashboard responsive with the available dataset size.
NFR-5	**Availability**	Keep the workbook accessible from its saved Tableau location; public access depends on publishing and internet availability.
NFR-6	**Scalability**	Organize worksheets and calculations consistently so the workbook can be maintained and extended as more records or categories are added.